

2] Explain the advantages of using stored procedures.

Ans:

Advantages of Using Stored Procedures

1. Improved Performance:

Stored procedures are precompiled and cached on the database server, reducing the need for repeated parsing and execution planning, which speeds up execution.

2. Reusability:

Once created, stored procedures can be reused multiple times across different applications or queries, saving development time and effort.

3. Reduced Network Traffic:

Multiple SQL statements are executed on the database server in a single call, reducing the need for multiple client-server interactions.

4. Enhanced Security:

Permissions can be granted on the stored procedure level instead of granting direct access to the underlying tables, ensuring data is accessed securely.

5. Maintainability:

Business logic can be centralized in the database as stored procedures, making it easier to update and maintain.

6. Modularity:

Complex operations can be broken into smaller, manageable stored procedures, improving code organization and readability.

7. Error Handling:

Stored procedures can include advanced error handling using control-flow statements like TRY...CATCH, enhancing robustness.

8. Control Flow Logic:

Stored procedures support loops, conditionals, and variables, allowing for more complex and dynamic operations than standard SQL queries.

9. Consistency:

Procedures ensure consistent implementation of business rules and operations, avoiding discrepancies in application logic.

10. Reduced Risk of SQL Injection:

Parameters in stored procedures are strongly typed, reducing the likelihood of SQL injection attacks compared to ad-hoc SQL queries.